plotly | Graphing Libraries (https://plotly.com/)(/graphing-libraries/)
m_medium=graphing_libraries&utm_content=sidebar)

R (/r) > 3D Charts (/r/3d-charts) > 3D Surface Plots

Suggest an edit to this page (https://github.com/plotly/plotly.r-docs/edit/master/r/2015-07-30-3d-surface-plots.Rmd)

Forum
(https://community.plotly.com/c/plotly-r-matlab-julia-net/29)

Pricing
(https://plotly.com/get-pricing/)

Dash
(https://dash.plotly.com)

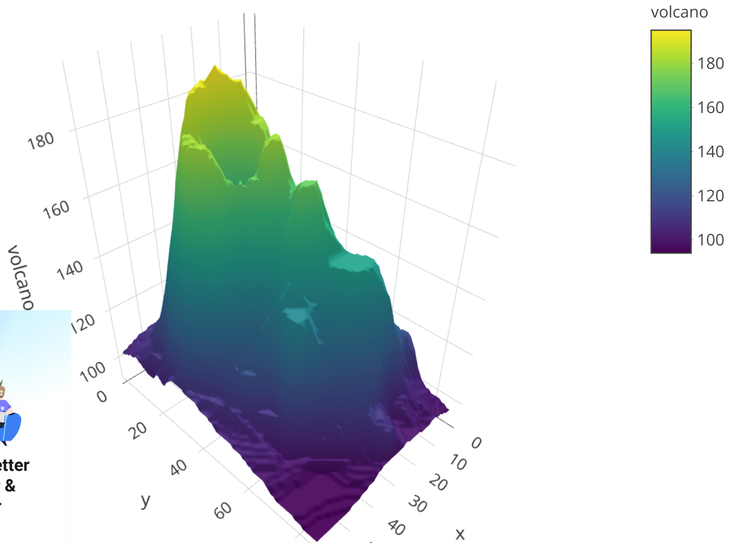
3D Surface Plots in R

How to make interactive 3D surface plots in R.

New to Plotly?

Basic 3D Surface Plot

```
library(plotly)
# volcano is a numeric matrix that ships with R
fig <- plot_ly(z = ~volcano)
fig <- fig %>% add_surface()

fig
```



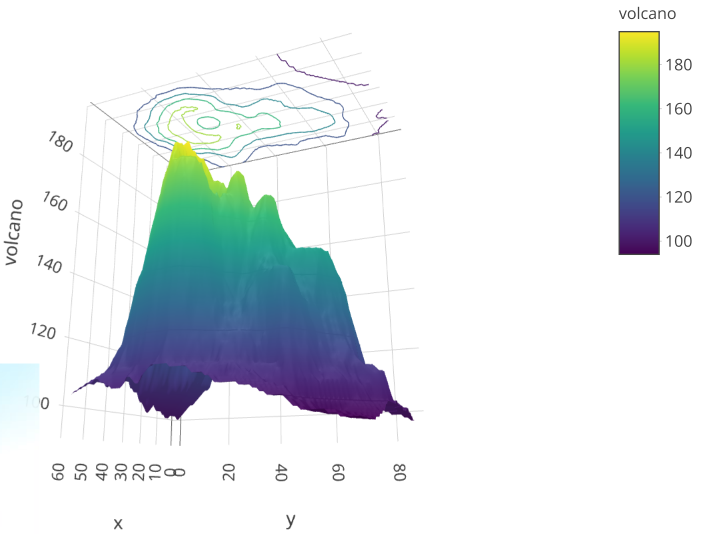
Surface Plot With Contours

```
library(plotly)
# volcano is a numeric matrix that ships with R
fig <- plot_ly(z = ~volcano) %>% add_surface(
  contours = list(
    z = list(
      show=TRUE,
      usecolormap=TRUE,
      highlightcolor="#ff0000",
      project=list(z=TRUE)
      )
    )
  )
fig <- fig %>% layout(
    scene = list(
      camera=list(
        eye = list(x=1.87, y=0.88, z=-0.64)
        )
      )
  )

fig
```

Forum
(https://community.plotly.com/c/plotly-r-matlab-julia-net/29)

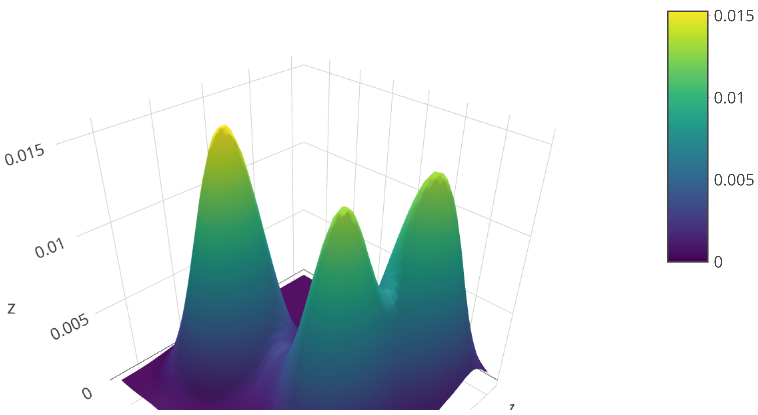
Pricing
(https://plotly.com/get-pricing/)

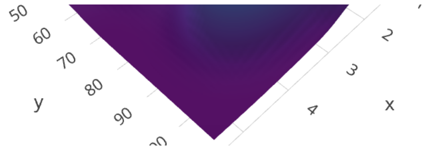
Dash
(https://dash.plotly.com)



2D Kernel Density Estimation

```
kd <- with(MASS::geyser, MASS::kde2d(duration, waiting, n = 50))
fig <- plot_ly(x = kd$x, y = kd$y, z = kd$z) %>% add_surface()

fig
```





Forum
(https://community.plotly.com/c/plotly-r-matlab-julia-net/29)

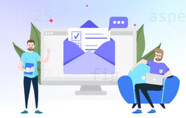
Pricing
(https://plotly.com/get-pricing/)

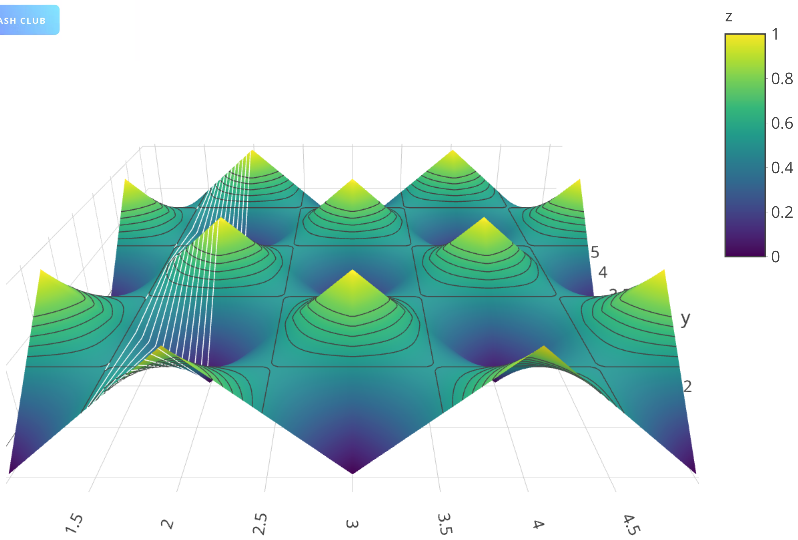
Dash
(https://dash.plotly.com)

Configure Surface Contour Levels

This example shows how to slice the surface graph on the desired position for each of x, y and z axis. contours.x.start (https://plotly.com/r/reference/#surface-contours-x-start) sets the starting contour level value, end sets the end of it, and size sets the step between each contour level.

```
x = c(1,2,3,4,5)
y = c(1,2,3,4,5)
z = rbind(
  c(0, 1, 0, 1, 0),
  c(1, 0, 1, 0, 1),
  c(0, 1, 0, 1, 0),
  c(1, 0, 1, 0, 1),
  c(0, 1, 0, 1, 0))


library(plotly)
fig <- plot_ly(
  type = 'surface',
  contours = list(
    x = list(show = TRUE, start = 1.5, end = 2, size = 0.04, color = 'white'),
    z = list(show = TRUE, start = 0.5, end = 0.8, size = 0.05)),
  x = ~x,
  y = ~y,
  z = ~z)
fig <- fig %>% layout(
    scene = list(
      xaxis = list(nticks = 20),
      zaxis = list(nticks = 4),
      camera = list(eye = list(x = 0, y = -1, z = 0.5)),
      aspectratio = list(x = .9, y = .8, z = 0.2)))
```





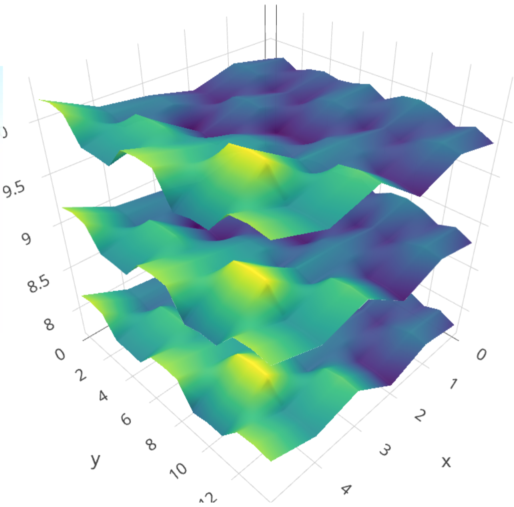
Multiple Surfaces

```r
z <- c(
  c(8.83,8.89,8.81,8.87,8.9,8.87),
  c(8.89,8.94,8.85,8.94,8.96,8.92),
  c(8.84,8.9,8.82,8.92,8.93,8.91),
  c(8.79,8.85,8.79,8.9,8.94,8.92),
  c(8.79,8.88,8.81,8.9,8.95,8.92),
  c(8.8,8.82,8.78,8.91,8.94,8.92),
  c(8.75,8.78,8.77,8.91,8.95,8.92),
  c(8.8,8.8,8.77,8.91,8.95,8.94),
  c(8.74,8.81,8.76,8.93,8.98,8.99),
  c(8.89,8.99,8.92,9.1,9.13,9.11),
  c(8.97,8.97,8.91,9.09,9.11,9.11),
  c(9.04,9.08,9.05,9.25,9.28,9.27),
  c(9,9.01,9,9.2,9.23,9.2),
  c(8.99,8.99,8.98,9.18,9.2,9.19),
  c(8.93,8.97,8.97,9.18,9.2,9.18)
)
dim(z) <- c(15,6)
z2 <- z + 1
z3 <- z - 1

fig <- plot_ly(showscale = FALSE)
fig <- fig %>% add_surface(z = ~z)
fig <- fig %>% add_surface(z = ~z2, opacity = 0.98)
fig <- fig %>% add_surface(z = ~z3, opacity = 0.98)

fig
```

Forum
(https://community.plotly.com/c/plotly-r-matlab-julia-net/29)

Pricing
(https://plotly.com/get-pricing/)

Dash
(https://dash.plotly.com)



What About Dash?

Dash for R (https://dashr.plot.ly/) is an open-source framework for building analytical applications, with no Javascript required, and it is tightly integrated with the Plotly graphing library.

Learn about how to install Dash for R at https://dashr.plot.ly/installation.

Everywhere in this page that you see fig , you can display the same figure in a Dash for R application by passing it to the figure argument of the Graph component (https://dashr.plot.ly/dash-core-components/graph) from the built-in dashCoreComponents package like this:

```r
library(plotly)

fig <- plot_ly()
# fig <- fig %>% add_trace( ... )
# fig <- fig %>% layout( ... )

library(dash)
library(dashCoreComponents)
library(dashHtmlComponents)

app <- Dash$new()
app$layout(
    htmlDiv(
        list(
            dccGraph(figure=fig)
        )
     )
)

app$run_server(debug=TRUE, dev_tools_hot_reload=FALSE)
```

Forum
(https://community.plotly.com/c/plotly
r-matlab-julia-net/29)

Pricing
(https://plotly.com/get-pricing/)

Dash
(https://dash.plotly.com)

Products

Dash (https://plotly.com/dash/)
Consulting and Training (https://plotly.com/consulting-and-oem/)

Pricing

Enterprise Pricing (https://plotly.com/get-pricing/)

About Us

Careers (https://plotly.com/company/careers)
Resources (https://plotly.com/resources/)
Blog (https://medium.com/@plotlygraphs)

Support

Community Support (https://community.plot.ly/)
Documentation (https://plotly.com/graphing-libraries)

JOIN OUR MAILING LIST

Sign up to stay in the loop with all things Plotly — from Dash Club to product updates, webinars, and more!

SUBSCRIBE (HTTPS://GO.PLOT.LY/SUBSCRIPTION)

Terms of Service (https://plotly.com/terms-of-service/) Privacy Policy (https://plotly.com/privacy/)